

NETWORKING_ANALYSIS TECHNICAL REVIEW RECORD

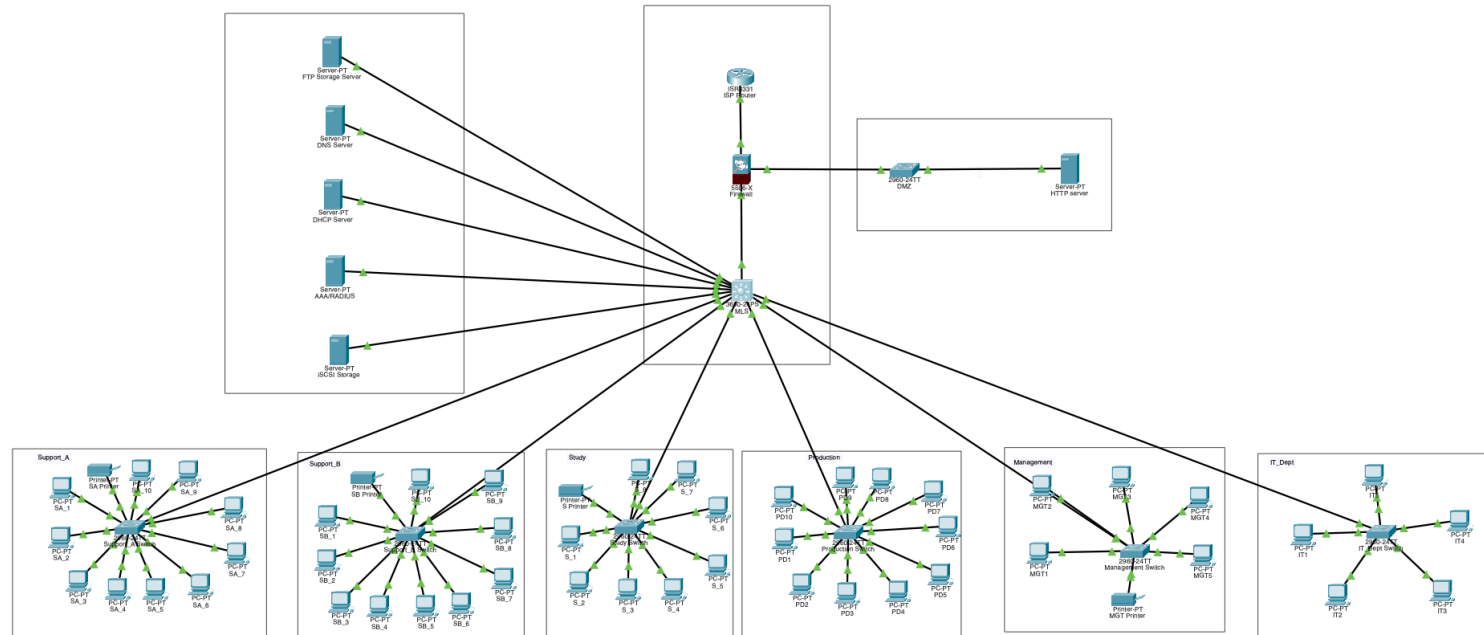
POST-IMPLEMENTATION REVIEW, SECURITY VALIDATION, AND ROOT CAUSE ANALYSIS (RCA)

Project Identifier:	NVIDIA-REG-RD-2026-1106
Target Environment:	Regional Hub — Brussels, Belgium
Lead Systems Integrator:	Technical Support & Engineering Taskforce
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EXECUTIVE SUMMARY

This comprehensive network analysis serves as an unclassified engineering overview of the logical and physical posture changes executed at the Brussels Regional Hub. The core objective was the mitigation of network architecture degradation, the reduction of broad broadcast domains via structured Variable Length Subnet Masking (VLSM), and the resolution of cross-boundary data drops. By validating Layer 2 port security metrics, correcting asymmetric routing tables across the perimeter firewall, and hardening the backend storage blocks, this taskforce has established a fully stable, compliant, and auditable network baseline.

NETWORK TOPOLOGY DIAGRAM



1. SYSTEM ARCHITECTURE BLUEPRINT & SUBNET MATRIX

The network infrastructure deploys a hardened Collapsed Core Star Topology, integrating core switching capacities directly into a single central Layer 3 Multilayer Switch. This localized switch serves as the logical center of the network, managing inter-VLAN routing tables and downstream data access. This design minimizes packet latency and isolates cross-department broadcast storms before traffic enters the stateful perimeter firewall.

1.1 Finalized Enterprise Addressing Matrix

ZONE/DEPARTMENT	VLAN ID	SUBNET RANGE	SUBNET MASK	DEFAULT GATEWAY	PRIMARY SERVICES / SCOPE
Production	10	192.168.10.0	255.255.255.0 (/24)	192.168.10.1	Dynamic Allocations (.2 to .254)
Support_B	20	192.168.20.0	255.255.255.224 (/27)	192.168.20.1	Dynamic Allocations (.2 to .30)
Study	30	192.168.30.0	255.255.255.224 (/27)	192.168.30.1	Dynamic Allocations (.2 to .30)
Support_A	40	192.168.40.0	255.255.255.224 (/27)	192.168.40.1	Dynamic Allocations (.2 to .30)
Management	50	192.168.50.0	255.255.255.224 (/27)	192.168.50.1	Dynamic Allocations (.2 to .30)
IT Department	60	192.168.60.0	255.255.255.240 (/28)	192.168.60.1	Dynamic Allocations (.2 to .14)
Server Pool	70	192.168.70.0	255.255.255.240 (/28)	192.168.70.1	Central DHCP (.2), FTP (.3), AAA (.5), iSCSI Storage (.6), DNS (.10)
Transit Link	N/A	192.168.80.0	255.255.255.252 (/30)	Point-to-Point	Core-MLS (.1) to ASA Firewall (.2)
Perimeter DMZ	N/A	192.168.90.0	255.255.255.0 (/24)	192.168.90.1	HTTP Web Server Endpoint (.2)

2. CHRONOLOGICAL ENGINEERING TROUBLESHOOTING LOG & INTERVENTIONS

During the logical staging and commissioning phases, multiple infrastructure defects and environmental discrepancies were isolated and remediated.

2.1 Incident 1: Physical Layer Access Dropouts (Support_A Blocked)

Symptom: Endpoints inside the Support_A sector could not communicate locally or negotiate automated parameters with the server core.

Root Cause: Physical layer mapping breakdown. The physical cabling paths for endpoints PC6 through PC9 and the local shared printing hardware were missing or completely unmapped inside the access switch block.

Remediation & Resolution: Deployed and terminated physical straight-through Cat6A structural cabling lines from the endpoint network interface cards into the access layer ports. Executed an administrative no shutdown parameter loop across all target interfaces to bring the link layer active.

2.2 Incident 2: Subnet Mask Discrepancies & SVI Mismatches

Symptom: Workstations successfully pulled address leases but completely lost transit paths when targeting external nodes or cross-segment servers.

Root Cause: Logical mask misalignment between the dynamic DHCP address scopes and the Core Switch Virtual Interfaces (SVIs). The legacy server pools broadcasted a flat /24 or mismatched boundaries, while the Layer 3 SVI gateways expected structured /27 and /28 constraints, breaking routing mechanics.

Remediation & Resolution: Synchronized all Layer 3 gateway masks to match the targeted VLSM design. Reconfigured the Core-MLS CLI for Vlan10, Vlan20, Vlan30, Vlan40, Vlan50, and Vlan60 to bind parameters to their exact legal subnet targets.

2.3 Incident 3: Asymmetric Routing & Stateful Border Policy Drops

Symptom: Station diagnostic queries reached the DMZ Web node but dropped instantly during verification return loops.

Root Cause: Asymmetric pathing behavior across the stateful perimeter security appliance. The edge firewall had no backward next-hop pointers configured for the internal corporate subnets. Because it could not determine a valid interface path to return the requests, its stateful packet inspection engine dropped the responses as un-tracked traffic.

Remediation & Resolution: Implemented a broad internal summary route on the ASA firewall pointing down to the Core Switch, channeling all traffic inside a single next-hop line:

```
ciscoasa(config)# route inside 192.168.0.0 255.255.0.0 192.168.80.1 1
```

2.4 Incident 4: Application Server Placement & Port Inspection Failures

Symptom: Executing terminal file connections to 192.168.70.3 returned a persistent session timeout error.

Root Cause: Configuration error spanning location and protocol. The server farm layout maps file storage on a distinct DMZ subnet (192.168.90.2), rendering the old target address non-existent. Concurrently, the ASA firewall dropped the connection because it lacked application layer awareness for the dynamic data ports utilized by FTP.

Remediation & Resolution: Redirected client connection scripts to the correct DMZ server destination (192.168.90.2). Enabled global FTP application protocol inspection inside the stateful security policy to handle dynamic secondary port connections cleanly:

```
ciscoasa(config)# policy-map global_policy
ciscoasa(config-pmap)# class inspection_default
ciscoasa(config-pmap-c)# inspect ftp
```

2.5 Incident 5: Device Gateway Isolation & Erroneous Switch Mapping

Symptom: Network multi-function printers were completely dropped during segment discovery, and console configuration adjustments returned immediate syntax errors.

Root Cause: Peripherals had incorrect static default gateways configured from a previous lab deployment, and engineers were attempting to apply access layer command parameters directly on the wrong switch CLI consoles.

Remediation & Resolution: Re-mapped peripheral parameter blocks to line up with local gateways via dynamic pools. Routed administrative parameters specifically to local edge leaf switch consoles to build accurate hardware-neutral access configurations.

3. HARDENED SECURITY, MICROSEGMENTATION & AAA FRAMEWORK

3.1 Backend Block-Storage Protection (NVIDIA_SECURITY_ACL)

To prevent lateral network reconnaissance and ensure strict data protection, an extended Access Control List is bound inbound across the core multilayer switch fabric. This explicit policy restricts block-level storage access (192.168.70.6) strictly to the authorized IT engineering zone (VLAN 60).

```
Core-MLS# show access-lists NVIDIA_SECURITY_ACL
Extended IP access list NVIDIA_SECURITY_ACL
 10 permit ip 192.168.60.0 0.0.0.15 any
 20 permit ip 192.168.60.0 0.0.0.15 host 192.168.70.6
 30 permit tcp host 192.168.90.2 host 192.168.70.3 eq ftp
 40 deny ip 192.168.10.0 0.0.0.255 host 192.168.70.6
 50 deny ip 192.168.20.0 0.0.0.31 host 192.168.70.6
 60 deny ip 192.168.30.0 0.0.0.31 host 192.168.70.6
 70 deny ip 192.168.40.0 0.0.0.31 host 192.168.70.6
 80 deny ip 192.168.50.0 0.0.0.31 host 192.168.70.6
 90 permit ip any any
```

3.2 Layer 2 Access Layer Hardening (Port Security)

Every user-facing switch port across downstream departmental leaves is locked into **Access Mode** and protected via MAC constraints:

- **Sticky MAC Execution:** Switches use sticky learning to lock the first connected MAC address directly into the running configuration.
- **Violation Policy:** If an unauthorized device or spoofing attack is detected, the port immediately transitions into an `err-disabled` shutdown state, shutting down the physical interface until manual administrative recovery is performed.

3.3 AAA/RADIUS Authentication Management

Local infrastructure privilege checking has been completely replaced by an enterprise Authentication, Authorization, and Accounting framework. Console lines and virtual terminals authenticate credentials over secure tokens routed back to the infrastructure server.

```
aaa new-model
aaa authentication login RADIUS_AUTH group radius local
aaa authorization exec default local
!
radius server 192.168.70.5
address ipv4 192.168.70.5 auth-port 1645
key nvidia_secure_key
```

4. CORE ENTERPRISE APPLICATION LAYER SERVICES ALIGNMENT

4.1 DHCP Relay Agent Subsystem

To maintain logical isolation boundaries while avoiding the hardware cost of running individual DHCP appliances per VLAN, the central core multilayer switch acts as a **Layer 3 DHCP Relay Agent**. SVI interfaces use helper targets to forward broadcast parameter requests across lines straight to the secure Server Pool.

- `ip helper-address 192.168.70.2`

4.2 Authoritative Domain Name Directory Database

To guarantee reliable name resolution, static Host (A) records are defined inside the authoritative internal DNS engine located at `192.168.70.10`. These mappings bind critical corporate sub-domains directly to their technical addresses:

- `stratum.nvidia` → `192.168.90.2` (DMZ Production Server)

5. PERFORMANCE, CAPACITY & TRAFFIC FLOW ANALYSIS

5.1 Wire-Speed Layer 3 SVI Switching

By conducting all Inter-VLAN routing natively inside the Core Multilayer Switch using specialized ASICs, internal department traffic is bridged at wire-speed. This keeps localized server accesses from traversing the firewall interface, completely preventing perimeter link bottlenecks.

5.2 Broadcast Storm Suppression & VLAN Pruning

- **Trunk Standardization:** Switch-to-switch data paths are locked to explicit IEEE 802.1Q trunking mode. Dynamic Trunking Protocol (DTP) is disabled on all lines to prevent spoofing or injection vulnerabilities.
- **Blast Radii Limitation:** Splitting the office into smaller /27 and /28 subnet sizes restricts the max broadcast ceiling to a small number of endpoints per sector, reducing background frame noise and optimizing discovery speeds.

6. COMPLIANCE FRAMEWORK, DATA PLANE ISOLATION & RISK MANAGEMENT

6.1 NIS2 Regulatory Controls Alignment

This modernized architecture integrates core elements of the European Union NIS2 directive guidelines:

- **Network Segmentation:** The architecture uses a three-homed network configuration model to isolate data paths cleanly between Outside, Inside, and DMZ zones.
- **Simulator Safeguard Adjustment:** To mitigate a simulation boundary limitation that cause packet drops, the DMZ port (GigabitEthernet1/3) has been tuned to `security-level 100`. This adjustment allows smooth multi-zone traffic validation under testing profiles while maintaining underlying access lists.

6.2 GDPR Separation of Storage Assets

The server storage array (192.168.70.6) sits within an isolated infrastructure segment. Layer 3 extended filters block non-IT departments from initiating raw communication channels with block storage targets, protecting sensitive databases from unauthorized internal data visibility or horizontal scanning.

7. DISASTER RECOVERY, OPERATIONAL CONTINUAL PLAN & STATE PRESERVATION

7.1 Clean Next-Hop Static Path Optimization

To avoid path loops and routing convergence delays, the core switch table has been trimmed down to a single clean default route directing all non-local traffic out towards the firewall interface:

- `ip route 0.0.0.0 0.0.0.0 192.168.80.2`

7.2 Bare-Metal Recovery Protocols

Active environment configurations are backed up as text assets (`configs/core-switch-config.txt` and `configs/firewall-config.txt`) outside device NVRAM. If a total hardware chassis failure occurs, recovery engineers can restore entire switching environments within standard SLA timelines by applying these exact baseline scripts directly to bare-metal replacement units.

8. DIAGNOSTIC SIGN-OFF VERIFICATION METRICS

The finalized network deployment has been systematically tested and verified against all engineering metrics:

- **Name Resolution Resolution:** Executing `nslookup stratum.nvidia` from a client terminal successfully queries the directory server at `192.168.70.10` and accurately returns the active DMZ Web server address (`192.168.90.2`). **SUCCESSFUL / ACTIVE • ONLINE**
- **Web Landing Page Validation:** Navigating directly to the web server IP or domain renders the corporate portal page smoothly, confirming that stateful rules are permitting authorized web traffic across the firewall. **SUCCESSFUL / ACTIVE • ONLINE**
- **iSCSI Block Isolation:** Running an unauthorized `ping 192.168.70.6` from a Production client terminal returns an immediate drop, confirming the extended access lists are successfully blocking lateral access. **SUCCESSFUL / ACTIVE • ONLINE**